

Ashish Gupta

High-Impact Technical Partner and Senior AI/ML & Embedded Engineer with 5.5+ years of experience architecting and deploying autonomous agentic AI systems, complex multi-agent workflows, scalable full-stack data platforms, and real-time embedded systems. Proven expertise in developing LLM-driven architectures (Llama-2, GEMMA, ChatGPT), advanced RAG pipelines, and LLM-based reasoning patterns (ReAct, Plan-Execute-Reflect) for enterprise automation. Highly skilled in Python, SQL, C++, and deploying cloud-native AI solutions (AWS/GCP, Docker, Kubernetes) with a strong focus on MLOps, CI/CD, engineering rigor, and operational resilience.

PROFESSIONAL EXPERIENCE

NEC software solutions, Mumbai : Senior Software Engineer – Dev-ops Automation

MAR 2026 - May 2026

Architecting and maintaining high-availability infrastructure using **Ansible** for automated orchestration, ensuring seamless deployments across distributed remote systems. Managing high-throughput data persistence with Cassandra and optimizing network traffic and security protocols via Nginx to maintain enterprise-grade service reliability.

Projects at NEC software solutions:

1. Automated Enterprise Deployment Pipeline – Engineered a comprehensive automation framework using **Ansible** to manage the end-to-end deployment of Java applications including single and multinode.

- Developed Ansible playbooks to automate **Nginx** server setup, Cassandra DB setups and **SSL** certificate management for JAVA application, securing all service traffic via HTTPS through a fully code-driven approach, user **Github Copilot** for AI-assisted Script development.
- Use **python** pytest for test
- Tech Stack:** JAVA, Ansible, Nginx, SSH, Cassandra, Linux, SSL/TLS, Github Copilot, Python, Pytest.

Snapwork, Mumbai : Senior Full Stack Software and Ai integration Engineer

SEPT 2024 - MAR 2026

Built scalable APIs, event-driven architectures, and **agentic AI** integrations using Node.js, Python, and C++. Mentored junior engineers in **AI engineering** best practices, modular development, and optimized AI model serving and deployments. improved frontend performance by **45%** using **React**, **TypeScript**, and **Webpack**, optimized deployments with custom shell scripts, and mentored junior developers in debugging, testing, and modular development. Utilize **Copilot** for **Flutter** app development.

Projects at Snapwork:

1. Julius Baer – Portfolio Management System (As Senior fullstack developer)

- Migrated and optimized large **Vue.js** components to **React.js** + **TypeScript** following performance & memory best practices.
- Analyzed and understood a legacy **C#/.NET** backend, then migrated it to a **Rust (Axum)** backend integrated with **C++** services, with a focus on native low-level performance and system efficiency.
- Successfully integrated an **LLM (GEMMA-3)** via the Hugging Face library to provide AI-powered customer support, establishing exception handling and **secure boundaries** within the platform.
- Tech Stack:** **Vue.js**, **React.js**, **TypeScript**, **Rust**, **C#/.NET**, **Python**, **Hugging face (GEMMA)**, **Rust (Axum)**, **C++**, **Webpack**, **MySQL**, **LLMs**

2. Medicard – Customer Support AI & Flutter Modules

- Architected and built an advanced **RAG (Retrieval-Augmented Generation)** pipeline for a Medicard customer support agent, optimized for agent memory, long-term context retention, and high retrieval accuracy.
- Developed agentic components to assist users, integrating **LLM** based reasoning and robust task delegation mechanisms.
- Developed new Flutter-based modules for the MediGo App using Copilot-assisted workflows.
- Researched and implemented Spec-Driven Development using **GitHub SpecKit**.
- Tech Stack:**, **SpecKit**, **Python**, **LLMs**, **RAG frameworks**, **Vector DBs**, **Flutter**, **Dart**, **Copilot**.

CODE-B, Mumbai : Full Stack Software, Ai/ML & Embedded Engineer → Senior Engineer

MAR 2021 - SEPT 2024

Designed high-performance microservices, multi-agent AI workflows, and distributed edge systems. Managed the end-to-end ML lifecycle from experimentation to deployment, building real-time ecosystems

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SOCIAL LINKS

- [Linked In](#)
- [Github](#)
- [HackerRank \(5 Stars\)](#)
- [X \(Twitter\)](#)

SKILLS

→ Languages

- ◆ C/C++
- ◆ C#/.NET
- ◆ Golang
- ◆ Rust
- ◆ Python
- ◆ JavaScript/TypeScript
- ◆ Java

→ Embedded & Edge

- ◆ ESP32 (Arduino) Bluetooth
- ◆ FreeRTOS
- ◆ Yocto Builder Project
- ◆ Embedded Linux
- ◆ IOT
- ◆ MQTT
- ◆ Raspberry Pi
- ◆ Shell scripting
- ◆ Pi Camera

→ AI/ML

- ◆ TensorFlow
- ◆ RAG Pipelines
- ◆ HIL Controls
- ◆ LLAMA (LLM), GEMMA(SLM), ChatGPT (OpenAi)
- ◆ Ollama
- ◆ Transfer Learning/Hugging Face
- ◆ Deep Learning
- ◆ NLP
- ◆ TensorRT
- ◆ Langchain
- ◆ OpenCV/Computer Vision

→ Cloud & DevOps

- ◆ AWS
- ◆ GCP
- ◆ Firebase
- ◆ Docker
- ◆ Cloud Run, Git
- ◆ SQL/MySQL/PostgreSQL
- ◆ MongoDB/Mongoose
- ◆ Cassandra
- ◆ Linux

using AWS/GCP, MERN stack, and lightweight edge inference tools. Designed ESP32 firmware in C++ and Rust, optimized sensor polling, and built Golang microservices with sub-100ms latency, along with Raspberry Pi I2C/GPIO drivers and lightweight edge inference tools integrating LLama-2 and other ML models. Developed a full-stack EMS (Energy/Inventory Management System) using AWS IoT Core (MQTT) and the MERN stack, enabling real-time power monitoring and remote device control via mobile and web applications.

Projects at Code-B:

- 1. Tech Agriculture** – AI-Powered Plant Care & Diagnostic System (As Senior fullstack and ML engineer)
 - Architected **multi-agent** workflows (planning, execution, and reflection loops) for a comprehensive AI diagnostic system using fine-tuned Llama-2 and TensorFlow (**ResNet50**)
 - Implemented LLM-based reasoning patterns (**ReAct**, **Plan-Execute-Reflect**) to orchestrate autonomous decision-making agents capable of dynamically generating remedy suggestions and care plans, deployed end-to-end on GCP.
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 - Tech Stack:** Python, TensorFlow, PyTorch, Multi-Agent Frameworks, ReAct, Llama-2, NLP, FastAPI, React.js, GCP.
- 2. Selec IoT** – IoT Device Inventory & Energy Monitoring System (As Senior Fullstack and IOT dev)
 - Delivered full **IoT** ecosystem with **React.js** Dashboard, **React Native** App, and **Node.js** backend for real-time monitoring, alerts, and monthly billing. Used graph-ql for past device data fetching using get device data function.
 - Developed real-time device firmware using Arduino C++/ESP32 governed by **FreeRTOS** for deterministic task scheduling. Built C++ and Rust microservices for high-speed parsing.
 - Implemented **real-time** data streaming using **Mqtt** and **AWS IoT** Core, ensuring stable device-to-cloud communication.
 - Debugged and integrated industrial protocols including **Modbus** for seamless edge telemetry.
 - Tech Stack:** AWS IoT, React.js (TS), GraphQL, React Native, Node.js, Rust, C++, Arduino C++, ESP32, Bootstrap, Docker, MongoDB.
- 3. Penrose Systems** – MegaMatch Multiplayer Match Game (As Senior Fullstack dev managed team)
 - Built real-time multiplayer game using Node.js/Nest.js, Socket.IO, Redis, and MongoDB, with **Vue.js** + **Pixi.js** for gameplay and animations, deployed on GCP.
 - Designed scalable event-driven architecture enabling fast matchmaking, synchronized player movement, and smooth gameplay.
 - Tech Stack:** Node.js, Nest.js, Socket.IO, Redis, Pixi.js, Vue.js, MongoDB, GCP.

Freelancing , Mumbai : *Software Engineer – Full Stack & Embedded Integration*

MAR 2020 - FEB 2021

Developed and deployed mobile-responsive MERN-stack dashboards on GCP Cloud Run, featuring clean UI/UX and seamless integration with microcontroller hardware via MQTT and REST APIs.

- Built dashboards and backend tools using MERN stack and deployed them on GCP Cloud Run.
- Delivered clean UI/UX and mobile-responsive applications.
- Integrated microcontroller-based solutions with web platforms using MQTT and REST APIs.

EDUCATION

Gurunanak College, University of Mumbai, Mumbai – *Bachelor of Science in Information Technology*

JUNE 2019 - JULY 2022

Completed a three-year **undergraduate** program combining core subjects in **mathematics**, **computer science**, and emerging technologies. The curriculum emphasized **software development**, **Embedded Systems**, **IOT**, **data structures**, **networking**, **databases**, **applied mathematics**, and practical lab-based problem solving.

ADDITIONAL PROJECTS

- **Pet Identity System (Pawfriend)** – Built ESP32 QR sync system for street animals using Python+ React + GCP.
- **Raspberry Pi Live Stream with Object Detection** – Implemented real-time video streaming with object detection using **OpenCV** and **Python** on Raspberry Pi. Created Custom grounded and smaller Linux build for Raspberry pi using **Yacto Project** deployed lightweight models to run at the edge with minimal latency. Used On device Image Classification using TensorFlow Lite.

- ◆ CI/CD Github Actions
- ◆ Firebase
- ◆ Nginx

→ Fullstack Web/Mobile Frameworks

- ◆ Node.js
- ◆ Express.js/Nest.js
- ◆ FastApi
- ◆ Flask
- ◆ GraphQL
- ◆ Vue.js/Nuxt
- ◆ PiXi.js
- ◆ React.js/Next.js/React Native

ACHIEVEMENTS

- **Fast-Track Promotion:** Promoted from Junior to Senior Software Engineer at Code-B within 2 years based on performance and mentoring contributions.
- **5-Star HackerRank Badge:** Earned 5-star rating in Problem Solving, showcasing strong algorithmic and analytical skills.
- **Media Recognition:** Pet Identity System project featured in media for its innovative approach to animal identification using ESP32 and cloud integration. [Media Link](#)

LANGUAGES

English – Full Professional Proficiency

Marathi – Intermediate

Hindi – Native